

A Counterexample to the BD-Polyhedral Characterisation of convex-DLD2P

Run `fa_banach_001`, lane 9

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Status

Candidate counterexample, likely valid. The packet answers negatively Problem 3.12 of Martin and Rueda Zoca [1], which asks whether a real L_1 -predual has the convex diametral local diameter two property (convex-DLD2P) if and only if it fails to be (BD) polyhedral.

Source Problem

The problem appears on page 12 of the source paper.

in the proof of Corollary 3.10. □

We do not know whether Corollary 3.9 or assertion (b) of Corollary 3.10 characterises the convex-DLD2P for L_1 -preduals. On the other hand, we also do not know if the convex-DLD2P for real L_1 -preduals can be characterised in terms of the failure of some kind of polyhedrality. Let us emphasise the question.

Problem 3.12. We do not know if a real L_1 -predual has the convex-DLD2P if and only if X fails to be (BD) polyhedral.

In the notation of the source, Δ_X denotes the set of Δ -points of S_X , and X has the convex-DLD2P when

$$B_X = \overline{\text{co}}(\Delta_X).$$

For a real Banach space X , property (BD) is

$$\sup\{x^*(x) : x^* \in \text{ext}(B_{X^*}) \setminus D(x)\} < 1 \quad (x \in S_X),$$

where $D(x) = \{x^* \in S_{X^*} : x^*(x) = 1\}$.

Counterexample

Let c be the real Banach space of convergent scalar sequences with the supremum norm. For $x = (x_n)_{n \geq 1} \in c$, write

$$x_0 = \lim_{n \rightarrow \infty} x_n.$$

Define

$$X = \{x \in c : 2x_0 = x_1 + x_2\}.$$

Proof Intuition

The source theorem characterises Δ -points in an L_1 -predual by looking at how infinitely many extreme dual functionals evaluate on a point. For this hyperplane the extreme dual functionals are just the coordinate functionals, so a point is a Δ -point exactly when its limiting coordinate has modulus one. But the equation $2x_0 = x_1 + x_2$ then forces $x_1 = x_2 = x_0$. Hence all Δ -points live in the closed convex slice $x_1 = x_2 = x_0$, while the unit ball also contains points with $x_1 \neq x_2$. At the same time, the point whose first two coordinates and limit are 1, but whose tail approaches 1 from below, witnesses failure of (BD).

Theorem 1. *The real Banach space $X = \{x \in c : 2\lim_n x_n = x_1 + x_2\}$ is an L_1 -predual, fails to be (BD) polyhedral, and does not have the convex-DLD2P.*

Proof. First we record that X is an L_1 -predual. In the notation of Casini, Miglierina, and Piasecki [2], let

$$f = \left(\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}, 0, 0, \dots\right) \in \ell_1,$$

where the first coordinate acts on $x_0 = \lim_n x_n$, and the following coordinates act on $x_1, x_2, \dots$. Then $\|f\|_1 = 1$ and

$$\ker f = \{x \in c : \tfrac{1}{2}x_0 - \tfrac{1}{4}x_1 - \tfrac{1}{4}x_2 = 0\} = X.$$

Since $|f_1| = 1/2$ and $|f_j| < 1/2$ for every $j \geq 2$, the hyperplane theorem of [2, Theorem 4.3] applies: the map

$$\phi : \ell_1 \longrightarrow X^*, \quad \phi(a)(x) = \sum_{n=1}^{\infty} a_n x_n,$$

is an onto isometry. Consequently X^* is isometric to ℓ_1 , and

$$\text{ext}(B_{X^*}) = \{\pm e_n^* : n \in \mathbb{N}\},$$

where $e_n^*(x) = x_n$.

We next identify the Δ -points of X . By [1, Theorem 3.2], for a real L_1 -predual Y and $u \in S_Y$, u is a Δ -point if and only if, for every $\varepsilon > 0$, the set

$$\{e^* \in \text{ext}(B_{Y^*}) : |e^*(u)| > 1 - \varepsilon\}$$

contains infinitely many pairwise linearly independent elements. In our space this says that $u \in S_X$ is a Δ -point if and only if infinitely many coordinates u_n are arbitrarily close in modulus to 1. Because $u \in c$, this is equivalent to

$$|u_0| = 1.$$

If $|u_0| = 1$, the constraint $2u_0 = u_1 + u_2$ and $|u_1|, |u_2| \leq 1$ force $u_1 = u_2 = u_0$. Hence

$$\Delta_X \subset M := \{x \in B_X : x_1 = x_2 = x_0\}.$$

The set M is closed and convex, so $\overline{\text{co}}(\Delta_X) \subset M$. However,

$$z = (1, -1, 0, 0, \dots) \in c, \quad z_0 = 0,$$

belongs to X , since $2z_0 = 0 = z_1 + z_2$, and $\|z\| = 1$. But $z \notin M$. Therefore

$$B_X \neq \overline{\text{co}}(\Delta_X),$$

so X does not have the convex-DLD2P.

It remains to check failure of (BD). Let $u \in X$ be given by

$$u_0 = 1, \quad u_1 = u_2 = 1, \quad u_n = 1 - \frac{1}{n} \quad (n \geq 3).$$

Then $u \in S_X$. For every $n \geq 3$,

$$e_n^*(u) = 1 - \frac{1}{n} < 1,$$

so $e_n^* \notin D(u)$. Nevertheless $e_n^*(u) \rightarrow 1$. Thus

$$\sup\{x^*(u) : x^* \in \text{ext}(B_{X^*}) \setminus D(u)\} = 1,$$

which is exactly the failure of (BD) at u . □

Verification Notes

The proof uses only two external facts:

- (1) the hyperplane duality theorem of Casini–Miglierina–Piasecki for $W_f \subset c$, applied to $f = (1/2, -1/4, -1/4, 0, \dots)$;
- (2) Theorem 3.2 of the source paper, characterising Δ -points in L_1 -preduals by extreme dual functionals.

The script `code/verify_counterexample.py` checks the displayed hyperplane equations and the elementary witnesses, but it is not used as a substitute for the proof.

Novelty Check

Before promotion, the run indexes were searched for 2010.15936, convex-DLD2P, BD polyhedral, Problem 3.12, and the explicit hyperplane pattern. No duplicate packet or attempt was found.

Web searches on June 18, 2026 for exact and close phrases, including “convex-DLD2P” with “BD polyhedral”, “Problem 3.12” with “convex-DLD2P”, and the explicit hyperplane pattern $2 \lim x = x_1 + x_2$, did not surface a later paper answering the problem or this construction.

Human Review Recommendation

The main review point is to confirm the exact indexing convention in [2, Theorem 4.3]. With the convention used there, the first coordinate of f multiplies $x_0 = \lim_n x_n$, and the example has $|f_1| = 1/2$, so the cited theorem applies.

References

- [1] M. Martin and A. Rueda Zoca, *Daugavet property in projective symmetric tensor products of Banach spaces*, arXiv:2010.15936.
- [2] E. Casini, E. Miglierina, and L. Piasecki, *Hyperplanes in the space of convergent sequences and preduals of ℓ_1* , arXiv:1410.7801.